CHAPTER 6 GENERAL DISCUSSIONS

The developments presented in this thesis introduce alternative numerical approaches at three distinct stages of the ice-accretion framework: Eulerian and Lagrangian formulations for droplet impingement, algebraic surface displacement and level-set methods for geometry evolution, and body-fitted and immersed-boundary approaches for the treatment of the evolving geometry. Each alternative introduces different compromises between computational efficiency, numerical robustness, implementation complexity, and modeling capabilities. This chapter discusses these trade-offs, with particular emphasis on how the appropriate methodology depends on the complexity and objectives of the icing simulation.

6.1 Eulerian and Lagrangian Droplet Impingement

The first two Ice Prediction Workshops primarily focused on assessing the ability of numerical icing tools to reproduce experimental ice shapes. Although the final ice geometry provides an important validation metric, ice accretion results from the successive coupling of the different modules discussed throughout this thesis. Agreement or disagreement in the final ice shape cannot therefore necessarily be attributed to a particular component of the icing framework. The recent third Ice Prediction Workshop (IPW3) consequently placed greater emphasis on the verification and validation of the individual components of the icing framework. For droplet impingement, one of the questions considered was whether the Eulerian and Lagrangian formulations converge toward consistent predictions despite their different descriptions of the dispersed phase.

The IPW3 test cases included the NACA0012 and the ONERA M6 wing, both considered in a 3D setup. The corresponding reference conditions prescribed for the workshop are summarized in Table 6.1. In the present section, the NACA0012 configuration is used to compare the integrated collected-water mass predicted by the Eulerian and Lagrangian formulations across successive grid refinements. The ONERA M6 configuration provides a complementary benchmark for assessing the computational performance of both formulations and the time required to obtain the corresponding impingement solutions.

Figure 6.1 compares the integrated collected-water mass predicted by the Eulerian and Lagrangian formulations over the grid sequence summarized in Table 6.2. Results are shown for droplets corresponding to the MVD of 20 μm. The difference between the two formulations remains below 2% over the complete grid sequence and systematically decreases with spatial

Table 6.1 Reference conditions for the IPW3 droplet-impingement benchmark cases.

	NACA0012	ONERA M6
M_∞	0.315	0.400
α [°]	4.0	2.0
Re	4.40×10^6	4.256×10^6
LWC [g/m^3]	0.55	0.51
MVD [μm]	20	20
c_{ref} [m]	0.5334	0.801673

Table 6.2 NACA0012 grid hierarchy.

Grid level	Elements
L4	5,861,000
L3	11,079,000
L2	26,988,000
L1	59,863,000

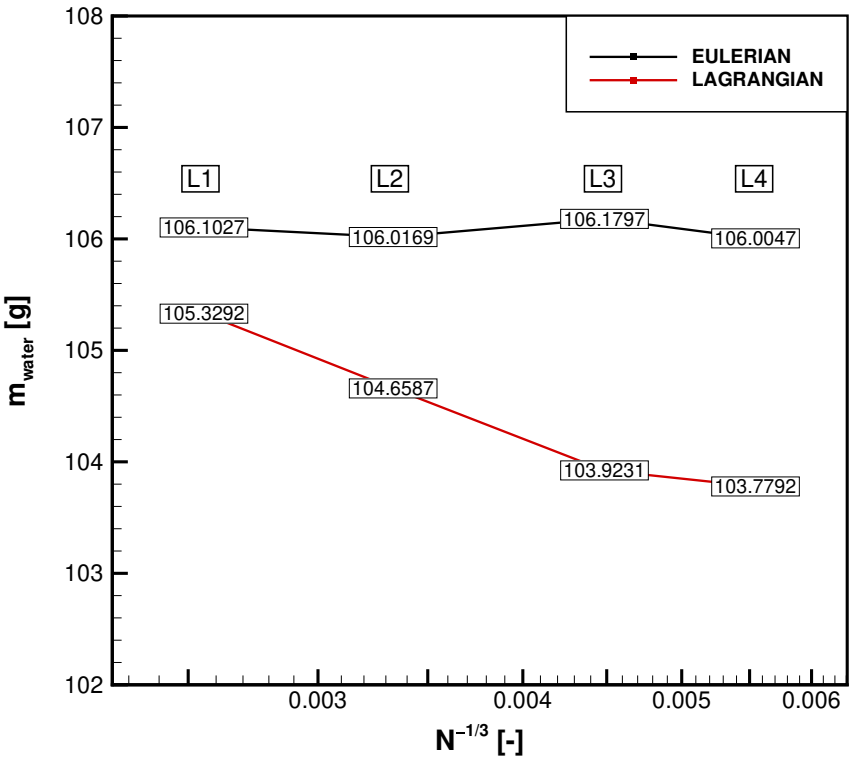


Figure 6.1 Spatial convergence of the integrated collected-water mass obtained with the Eulerian and Lagrangian droplet formulations for the NACA0012 IPW3 configuration using the MVD of 20 μm.

refinement. The figure also highlights a greater sensitivity of the Lagrangian prediction to the underlying volume-mesh resolution. This behavior can be attributed in part to the repeated interpolation of the carrier-flow quantities along a large number of droplet trajectories following different paths through the computational domain. Consequently, changes in the spatial resolution of the underlying flow field can influence the integrated collected-water mass through the trajectory calculations. In contrast, the Eulerian droplet equations are solved directly on the volume mesh using the local flow solution, resulting in considerably less variation of the integrated collected-water mass over the present grid sequence. Nonetheless, the Eulerian and Lagrangian formulations converge toward consistent integrated impingement predictions with spatial refinement, reaching a difference of 0.8% on Level 1.

The computational performance of the two formulations is compared for the ONERA M6 configuration, with representative timings reported in Table 6.3. The Level 4 calculation was performed on a workstation equipped with an Intel Xeon w7-2595X processor with 26 physical cores. The Level 1 calculation was performed on 8 Trillium cluster compute nodes, each containing two AMD EPYC 9655 processors with 96 cores per socket, corresponding to 192 physical cores per node. The hardware information is provided for reference to distinguish between the shared-memory and distributed-memory calculations.

Table 6.3 Representative computational performance of the Eulerian and Lagrangian droplet formulations for the ONERA M6 configuration.

Grid	Elements	Eul. [s]	Lag. [s]	ETR* [%]	Hardware
L4	135,168	83.05	15.40	81.5	Xeon w7-2595X
L1	69,206,016	1501.81	1182.13	21.3	8×2 AMD EPYC 9655

Note: ETR: execution-time reduction.

The Lagrangian formulation is particularly attractive when the computational mesh can be efficiently accessed within shared memory, allowing trajectories to traverse the domain without requiring remote access to flow and mesh data. For the ONERA M6 calculations reported here, the Lagrangian formulation reduces the execution time relative to the Eulerian formulation by 81.5% on Level 4 and 21.3% on Level 1. For the larger distributed-memory calculation, the required flow and mesh data are replicated across the compute nodes to minimize communication during trajectory integration. This replication represents a necessary trade-off: although it substantially reduces remote data accesses, it increases the memory requirements on each compute node and can become restrictive as the mesh size increases. In addition, the irregular memory-access patterns associated with trajectory tracking make the Lagrangian formulation more susceptible to memory-bandwidth limitations and cache misses as the number of concurrent trajectories increases. These effects reduce the computational

advantage of the Lagrangian formulation on the Level 1 case despite limiting inter-node communication.

The Eulerian formulation naturally follows the spatial decomposition of the computational mesh, making its parallel execution relatively straightforward to configure. In contrast, the Lagrangian performance reported here was achieved after careful optimization of the adaptive seeding strategy, trajectory workload balancing, and execution configuration according to the underlying cluster architecture. Efficient Lagrangian simulations on large 3D problems therefore require greater implementation effort and knowledge of the computing environment, while the Eulerian formulation is generally easier to configure for distributed-memory calculations. Ultimately, the choice between the two formulations depends on the computational scale of the problem and the flexibility required to represent the underlying droplet physics. In particular, the Lagrangian formulation offers greater flexibility for modeling complex SLD phenomena, such as splashing and secondary-droplet re-impingement, whereas the Eulerian formulation becomes particularly attractive for large-scale simulations when memory availability becomes a limiting factor.

6.2 Algebraic and Level-Set Ice Accretion

Explicit algebraic surface displacement presents important limitations for 3D multilayer ice accretion as increasingly complex ice features develop. Figure 6.2 compares the algebraic and level-set approaches under the same icing conditions and using the same number of accretion layers ($N_{\text{layers}} = 5$), such that the differences observed between the final geometries originate primarily from the geometry-evolution procedure. With the algebraic approach, successive displacement of the surface nodes causes the mesh to become increasingly susceptible to local folding and element entanglement. Extensive surface smoothing is therefore required between successive accretion steps to maintain a valid computational surface. Although this stabilizes the geometry-evolution procedure, it also progressively damps the smaller-scale ice features generated during the accretion process. This effect is clearly visible in Fig. 6.2, where the algebraic solution exhibits a considerably smoother and more uniform ice structure. In contrast, the level-set formulation allows the interface to evolve without being directly constrained by the connectivity of the previous surface mesh and requires little to no surface smoothing, thereby preserving substantially more of the 3D structures that develop during successive accretion layers.

This difference has implications beyond the prediction of an individual ice shape. In practice, the robustness of the geometry-evolution procedure determines which numerical experiments can be performed reliably. With algebraic displacement, increasingly complex geometries

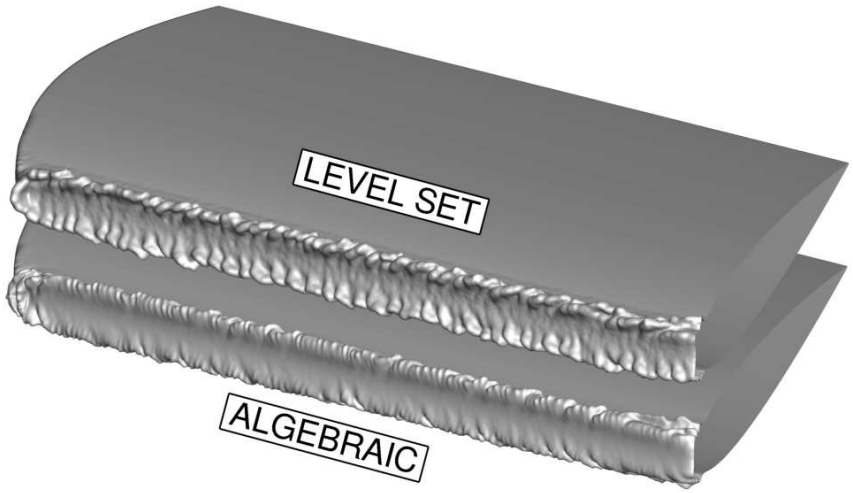


Figure 6.2 Comparison of algebraic and level-set ice shapes under identical icing conditions after five accretion layers.

or a larger number of accretion layers can require case-dependent smoothing and remeshing parameters, while severe surface distortion may still lead to failure before completion of the simulation. Consequently, the geometry-update procedure itself can become a limiting factor when attempting to investigate the sensitivity of the complete icing framework to physical and numerical parameters.

The level-set methodology substantially reduces this limitation and therefore provides a more suitable framework for systematic investigations of 3D multilayer ice accretion. Once the geometry evolution is sufficiently robust, different icing conditions, droplet-size distributions, spatial resolutions, and numbers of accretion layers can be investigated without the same concern that the evolving surface will fold onto itself or cause the simulation to fail midway through the accretion sequence. This capability is particularly important because the resulting 3D ice morphology can exhibit a strong sensitivity to these parameters. As illustrated in Fig. 6.3, different structures can emerge along the iced leading edge depending on the droplet-size distribution, the number of accretion layers, and the spatial resolution used to represent the evolving interface. The level-set methodology therefore provides not only a more robust geometry-evolution technique, but also an enabling tool for exploring the underlying numerical and physical mechanisms governing the development of complex 3D ice structures, such as scallop formation.

The level-set formulation nevertheless introduces a considerably more elaborate geometry-evolution procedure. Rather than directly displacing the surface mesh, the ice–air interface

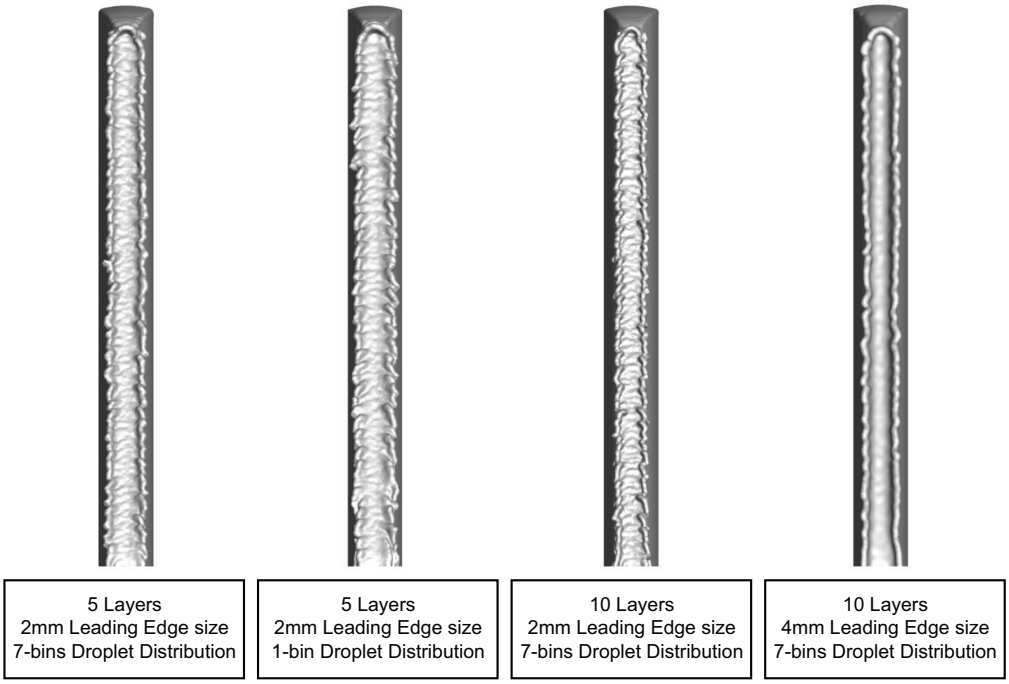


Figure 6.3 3D level-set ice shapes obtained for different numerical and physical configurations.

is represented implicitly as the zero contour of a level-set function defined over an auxiliary volume mesh. Each geometry update therefore requires the construction of this volume mesh, initialization of the signed-distance field, propagation of the required surface quantities into the volume, advection of the level-set field, and extraction of the updated zero-level-set surface. These additional operations increase both the computational cost and the complexity of setting up a simulation compared with direct algebraic displacement. Much of this workflow can, however, be automated, such that the additional complexity does not necessarily translate into a comparable increase in user intervention. The level-set methodology therefore represents a worthwhile trade-off for multilayer ice accretion. For single-layer calculations, however, algebraic node displacement is generally less susceptible to the geometric difficulties associated with successive surface updates and remains a simpler and computationally efficient alternative.

6.3 Body-Fitted and Immersed-Boundary Approaches

As CFD applications are extended toward increasingly complex geometrical configurations, mesh generation can represent a substantial portion of the overall simulation workflow. While generating an initial computational mesh around a complex geometry is often relatively

straightforward, obtaining a high-quality body-fitted mesh with adequate near-wall resolution and controlled element quality can require significant effort. This difficulty becomes particularly important for applications involving evolving geometries, such as ice accretion, where the computational mesh must accommodate increasingly complex ice surfaces over successive accretion layers.

Body-fitted meshes provide direct control of the near-wall resolution and are well suited to aerodynamic and icing calculations, for which wall shear stress and heat transfer depend strongly on the near-wall solution. Their main limitation for multilayer ice accretion is the need to maintain a high-quality geometry-conforming mesh as the ice surface evolves. As increasingly complex ice features develop, repeated deformation or regeneration of the volume mesh can therefore become a significant constraint on the robustness of the complete simulation.

Immersed-boundary methods alleviate this constraint by embedding the evolving geometry within a non-conforming background mesh, thereby avoiding repeated generation of a body-fitted RANS mesh. This flexibility comes at the expense of the spatial resolution required near the immersed interface. Accurate representation of the geometry and reconstruction of near-wall quantities require sufficiently small cells around the surface, and the background mesh must remain adequately refined as the ice grows away from the clean geometry.

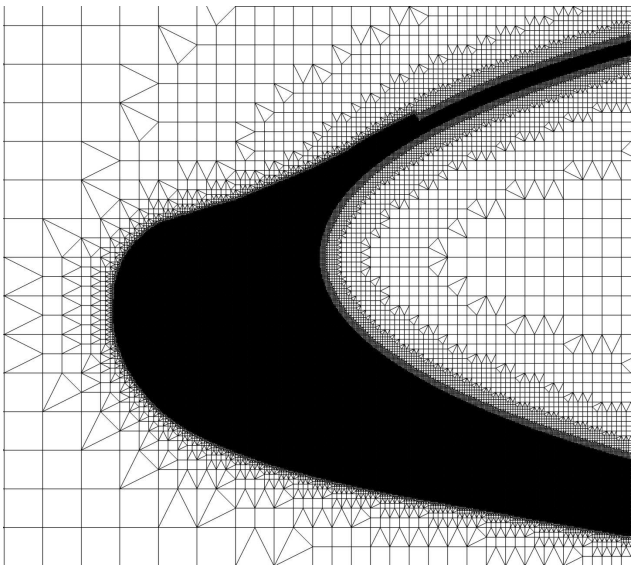


Figure 6.4 Example of the prescribed refined region surrounding the NACA0012 geometry and the anticipated ice-accretion region.

In Chapter 5, this requirement was addressed by prescribing a sufficiently large refined region around the anticipated ice-accretion region, as illustrated in Figure 6.4. While suitable for

the cases considered, this approach requires prior knowledge or an estimate of the expected ice geometry and can introduce unnecessary cells in regions where fine resolution is ultimately not required.

Adaptive mesh refinement provides a more general solution by allowing the background mesh to follow the evolving immersed interface while retaining coarser cells elsewhere. Combining immersed-boundary methods with efficient adaptive refinement would therefore preserve their principal advantage for multilayer icing—avoiding repeated body-fitted volume-mesh generation—without requiring a large predefined region of fine resolution. The main challenge is to perform this adaptation efficiently enough that its computational cost does not offset the advantages of the immersed-boundary approach.

The immersed-boundary formulation therefore provides an attractive alternative to body-fitted approaches for multilayer ice accretion, particularly when complex 3D ice geometries make repeated aerodynamic mesh generation increasingly difficult. However, its accuracy remains dependent on the treatment of the immersed interface and the reconstruction of near-wall quantities. The differences observed in the heat-transfer predictions under glaze conditions highlight the need for further improvements to the thermal wall treatment.